

1. Improving student retention with data driven analysis [Use Excel with graph]

Step-by-Step Process: Creating the Excel Dashboard

This section provides a detailed guide for building an interactive dashboard in Microsoft Excel using your student data.

Phase 1:

Data Preparation in Excel

1. Open Excel & Import Data:

- Open a new Excel workbook.
- Copy all your student data (including the header row) directly from your source.
- Paste the data into Sheet1 of your Excel workbook, starting from cell A1.

2. Convert Data to an Excel Table:

- Click on any single cell within your pasted data.
- Go to the **Insert** tab on the Excel ribbon.
- Click on the **Table** icon.
- In the "Create Table" dialog box, ensure that the selected range covers all your data and that "My table has headers" is checked. Click **OK**.
- **Rename Your Table:** With any cell in your new table selected, go to the **Table Design** tab (which appears contextually). In the "Table Name" field (usually on the far left), type a descriptive name like StudentData and press Enter. This makes it easier to reference your data in PivotTables.

Phase 2:

Creating Pivot Tables and Pivot Charts for Dashboard Elements

For each visual element you want on your dashboard, you will typically create a dedicated PivotTable and PivotChart on a separate sheet for organization.

1. Overall Student Retention (Dropout Rate):

- **Goal:** Visualize the percentage of students who dropped out.
- **Steps:**
 - Click any cell in your StudentData table (Sheet1). Go to **Insert > PivotTable**. Choose "New Worksheet" and click **OK**.
 - Rename this new sheet Retention_Overview.
 - In the PivotTable Fields pane:
 - Drag Dropped_Out to the 'Rows' area.
 - Drag Dropped_Out to the 'Values' area.
 - Right-click on the " **Overall Student Dropout Rate**"
 - in the Values area, select "Show Values As" > "% of Grand Total".
 - **Create Chart:** With the PivotTable selected, go to **PivotTable Analyze > PivotChart**. Select a **Pie Chart** or **Doughnut Chart**.

- **Refine Chart:** Add Data Labels (showing percentage) and a clear title like "Overall Student Dropout Rate."

2. Academic Performance (Final Grade Distribution):

- **Goal:** Show the distribution of final grades.
- **Steps:**
 - Create a New PivotTable on a new sheet (rename to Grades_Distribution).
 - Drag Final_Grade to 'Rows'. Drag Final_Grade to 'Values' (Count).
 - Group by Final grade
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 - **Create Chart:** Select a **Column Chart**.
 - **Refine Chart:** Title "Final Grade Distribution," label axes (e.g., "Grade," "Number of Students").

3. Impact of Class Failures on Retention:

- **Goal:** Show how the number of class failures correlates with dropout status.
- **Steps:**
 - Create a New PivotTable on a new sheet (rename to Failures_Impact).
 - Drag Number_of_class_Failures to 'Rows'.
 - Drag Dropped_Out to 'Columns'.
 - Drag Dropped_Out to 'Values' (Count).
 - **Create Chart:** Select a **clustered Column Chart**.
 - **Refine Chart:** Title "Dropout Status by Class Failures," label axes. This is a critical visual for retention.

4. Influence of Study Time on Grades:

- **Goal:** See if more study time leads to higher average grades.
- **Steps:**
 - Create a New PivotTable on a new sheet (rename to Study_Time_Impact).
 - Drag Study_Time to 'Rows'.
 - Drag Final_Grade to 'Values' (change 'Sum' to 'Average' via Value Field Settings).
 - Format number to 0 decimals
 - **Create Chart:** Select a **Pie of a Pie chart**
 - **Refine Chart:** Title "Average Final Grade by Study Time."

5. Effectiveness of Support Systems (School & Family Support):

- **Goal:** Compare average grades based on support received.
- **Steps:** (Repeat for School_Support and Family_Support)
 - Create a New PivotTable on a new sheet (rename to Support_Effect).
 - Drag Family_Support to 'Rows'.

- Drag School_Support to columns
- Drag Final_Grade to 'Values' (Average).
- **Create Chart:** Select a **3D stacked Bar chart**.
- **Refine Chart:** Title "Average Final Grade by School Support" (or "Family Support").

6. Absences and Academic Performance:

- **Goal:** Visualize the relationship between absences and average grades.
- **Steps:**
 - Create a New PivotTable on a new sheet (rename to Absences_Impact).
 - Drag Number_of_Absences to 'Rows'.
 - Drag Final_Grade to 'Values' (Average).
 - **Create Chart:** Select a **Line Chart**. Consider grouping the Number_of_Absences field if there are too many distinct values (right-click on the row label in PivotTable > Group).
 - **Refine Chart:** Title "Average Final Grade by Number of Absences."

7. Demographic Performance (e.g., Gender vs. Final Grade/Dropout):

- **Goal:** Understand if specific demographic groups perform differently.
- **Steps:**
 - Create a New PivotTable on a new sheet (rename to Demographics).
 - Drag Gender to 'Rows'.
 - Drag Final_Grade to 'Values' (Average).
 - **Create Chart:** Select a **Bar Chart**.
 - **Refine Chart:** Title "Average Final Grade by Gender."
 - Drag Gender to 'Rows'.
 - (You might create a separate one for Dropped_Out Count/% for clarity).
 - **Create Chart:** Select a **Bar Chart**.
 - **Refine Chart:** Title "Dropped_out Rate by Gender."

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Phase 3: Assembling the Dashboard

1. Create Your Dashboard Sheet:

- Insert a new blank sheet in your Excel workbook.
- Rename this sheet to Dashboard.
- Go to view and remove Gridlines

2. Copy and Paste Charts:

- Go to each of the sheets where you created your Pivot Charts (Retention_Overview, Grades_Distribution, etc.).
- Click on the Pivot Chart to select it.

- Copy it (Ctrl+C).
- Go to your Dashboard sheet and paste it (Ctrl+V).
- Repeat this for all the charts you want to display on your dashboard.

3. **Arrange and Resize Charts:**

- Carefully arrange the pasted charts on the Dashboard sheet.
- Resize them to fit the layout logically and ensure readability. Group related charts together.

Phase 4: Adding Interactivity with Slicers

1. **Insert Slicers:**

- Click on *any* Pivot Chart on your Dashboard sheet.
- Go to the **PivotChart Analyze** tab (or simply **Analyze** for newer Excel versions).
- Click on **Insert Slicer**.
- A dialog box will appear showing all your data fields. Select the fields you want to use as interactive filters for your dashboard. Good choices for retention analysis include:
 - Gender
 - Age
 - Address
 - Mother_Education
 - Father_Education
 - Internet_Access
 - School_Support
 - Family_Support
 - Extra_Curricular_Activities
 - Wants_Higher_Education
- Click **OK**. Your Slicers will appear on the Dashboard sheet.

2. **Connect Slicers to All Pivot Tables: This is a crucial step for interactivity.**

- Right-click on each Slicer you've added.
- Select **Report Connections...** (or "Connections..." in older versions).
- In the "Report Connections" dialog box, you will see a list of all PivotTables in your workbook. Check the box next to *every* PivotTable that you want this Slicer to control.
- Repeat this for every Slicer on your dashboard. Now, clicking an item on a Slicer will update all connected charts simultaneously.

Phase 5: Dashboard Design and Final Touches

1. **Add a Main Title:**

- Use a large font or a WordArt object (Insert > WordArt) to create a prominent title for your dashboard at the top (e.g., "Student Retention & Success Dashboard").

2. **Clean Up the View:**

- Go to the **View** tab on the ribbon and uncheck "Gridlines" and "Headings." This gives your dashboard a professional, clean look.

3. Format Charts and Slicers:

- **Chart Titles:** Ensure all charts have clear, concise titles that describe their content.
- **Axis Labels:** Make sure all chart axes are properly labeled with units if necessary.
- **Data Labels:** Add data labels to charts where appropriate (e.g., percentages on pie charts, specific values on bar charts).
- **Consistent Styling:** Use Excel's Chart Design options (under the "Design" tab when a chart is selected) to choose consistent colors and styles across all your visuals. You can also format Slicers using the "Slicer" tab.

4. **Add Explanatory Text (Optional):** Use text boxes (Insert > Text Box) to add brief descriptions or key takeaways next to complex charts.

5. **Test the Dashboard:** Interact with all the Slicers and ensure that the charts update as expected. Check for any errors or inconsistencies.

Output :-

